Memory Management for AI PA Agents

AI agents wake up fresh every session. Without deliberate memory management, they repeat mistakes, forget context, and frustrate the people they work with.

This document covers how to build a memory system that actually works.

The Core Problem

Every session starts from zero. The agent has no recollection of yesterday's conversation, last week's decision, or the correction you gave it three times already.

The solution isn't to load everything into context — that's expensive and slow. The solution is **structured, tiered memory**: the right information, in the right place, loaded at the right time.

Three Memory Tiers

Tier 1 — Working Memory (In-session)

Everything the agent sees in the current conversation. Loaded automatically.

- System prompt (SOUL.md, AGENTS.md, USER.md)
- MEMORY.md (long-term facts)
- Today's daily notes
- The conversation itself

Rule: Keep this under ~10,000 tokens total. Every token here costs money on every message.

Tier 2 — Session Memory (`memory/YYYY-MM-DD.md`)

A daily log of what happened. Written at the end of significant conversations.

Write: tasks assigned, decisions made, corrections received, context needed for tomorrow.
Skip: casual greetings, short acks, transient task state.

Rule: Write once per conversation (at the end), not after every message.

Tier 3 — Long-term Memory (`MEMORY.md`)

The agent's permanent memory. Curated, not raw.

Belongs here: owner preferences, key contacts, cross-session rules, lessons from mistakes.
Doesn't belong: completed one-time tasks, outdated rules, temporary status.

Rule: Promote to MEMORY.md only when a pattern appears 2+ times, or after an explicit correction.

Two Memory Types

FACT Something the owner stated directly.
"I work until 20:00"

DEDUCED A logical conclusion inferred from behavior or patterns.
"Prefers execution over explanation — never asks me to explain my reasoning, only wants results"

Write [DEDUCED] entries proactively. If you notice a pattern — capture it. Don't wait for the owner to state it explicitly.

WhatsApp Memory

Every DM conversation needs its own context file — incoming and outgoing.

```
memory/whatsapp/dms/<PHONE>/context.md
memory/whatsapp/groups/<JID-sanitized>/context.md
```

Critical rule: After sending any proactive message, create the context file immediately. If someone replies a day later in a new session, you'll have zero context without it.

What to log: who this person is, why you contacted them, what was decided, current status.

Preventing Bloat

Memory files grow over time. Left unchecked, they consume too much context window.

File	Target max
MEMORY.md	175 lines
AGENTS.md / system files	60 lines

Weekly compaction cron: Every Sunday, a lightweight model reads the memory files, removes outdated entries, merges duplicates, and pushes to git. No human involvement needed.

Signs you need compaction: MEMORY.md exceeds 200 lines · loading context that's no longer relevant · session startup feels heavy.

What to Store Where

Type	Location
Rules, preferences, key contacts	MEMORY.md
Daily events, task logs	memory/YYYY-MM-DD.md
WhatsApp conversation context	memory/whatsapp/dms or groups
Research, strategy, docs	monday.com
Credentials, config	Local files only (never monday)
Skill-specific config	skills/<name>/.context

The Compaction Mindset

Think of memory like a human's long-term memory — not a log file.

A human doesn't remember every conversation word for word. They remember what mattered, what they learned, who they can trust, and what decisions were made and why.

Your MEMORY.md should read the same way. **Curated, dense, useful** — not a transcript.

Quick Reference

Action	When
Write to daily notes	After any significant conversation (5+ exchanges)
Write to MEMORY.md	After 2+ repetitions, or after correction
Write WhatsApp DM context	Immediately after any message sent or received
Run compaction	Weekly (automated) or when >200 lines
Push to git	After any memory write

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Based on production patterns from the monday.com PA network